NOT MEASUREMENT SENSITIVE

A-A-59743
June 30, 2003

COMMERCIAL ITEM DESCRIPTION

CLEANING COMPOUND SOLVENT, DICHLOROPENTAFLUOROPROPANE (HCFC-225)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers a dichloropentafluoropropane (HCFC-225) cleaning compound solvent for precision cleaning applications.

2. CLASSIFICATION. The cleaning compound solvents are of the following types:

Type I - HCFC-225M cleaning compound solvent (see table I).
Type II - HCFC-225G cleaning compound solvent (see table I).

3. SALIENT CHARACTERISTICS

3.1 Material. HCFC-225 cleaning compound solvents shall be mixtures of 1,3 Dichloro-1,1,2,2,3, pentafluoropropane (HCFC-225cb) (Chemical Abstract Service [CAS] Registry Number 507-55-1), and 3,3 Dichloro-1,1,1,2,2, pentafluoropropane (HCFC-225ca) CAS Number 422-56-0).

3.2 Composition. HCFC-225 cleaning compound solvents shall be nonflammable, evaporating, non-residue solvents as specified in table I.

TABLE I. Composition and purity requirements.

Component	Component purity (percent by weight)	HCFC-225M Composition (percent by weight)	HCFC-225G Composition (percent by weight)
1,3 dichloro-1,1,2,2,3, pentafluoropropane (CAS No. 507-55-1)	See table II	50 - 60	99 - 100
3,3 dichloro-1,1,1,2,2, pentafluoropropane (CAS No. 422-56-0)	See table II	40 - 50	0 - 1

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100B120-3, Highway 547, Lakehurst, NJ 08733-5100.

FSC 6850

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3.3 Chemical and physical properties. HCFC-225 cleaning compound solvents shall conform to the requirements of table II, when tested in accordance with table II.

TABLE II. Chemical and physical properties.

Properties	HCFC-225M requirements	HCFC-225G requirements	Test method or paragraph 1/
Appearance	Clear, water white	Clear, water white	3.5
Boiling point at 760 millimeters barometric pressure	54.0 ±2.0 °C (129.0 ±3.6 °F)	56.1 ±2.0 °C (133.0 ±3.6 °F)	ASTM-D1120
Chemical purity	99.5 percent by weight minimum (min)	99.5 percent by weight (min)	ASTM-D3447
Impurities	225 derivatives	225 derivatives	ASTM-D3447
Moisture content	50 parts per million (ppm) by weight maximum (max.)	50 parts per million (ppm) by weight (max.)	ASTM-D3401
Free chlorine content	1.0 ppm by weight, max.	1.0 ppm by weight, max.	ASTM-D3443
Free fluoride content	1.0 ppm by weight, max.	1.0 ppm by weight, max.	ASTM-D3443
Non-volatile residue	2.0 ppm, by weight, max.	2.0 ppm, by weight, max.	ASTM-D2109 or ASTM-F331
Particle matter number of particles equal to or greater than 5.0 microns (µm) number of particles greater than or equal to 1.0 µm number of particles equal to or greater than 0.5 µm	3 per milliliter (mL), max. 5 per mL, max. 16 per mL, max.	3 per milliliter (mL), max. 5 per mL, max. 16 per mL, max.	ASTM-F661 or SAE-ARP598

1/ Manufacturer test procedures equivalent to those specified may be substituted.

3.4 Shelf life. HCFC-225 cleaning compound solvents specified in table I shall meet the specified requirements in table II after storage in unopened containers for not less than one year.

3.5 Appearance. HCFC-225 cleaning compound solvents shall be clear, colorless liquids (water white) with no evidence of particles. The appearance shall be determined by visually examining 500 ±50 mL of solvent for clarity, color, and solids in a clean, clear one-liter container. The inspection shall be performed by a person with normal visual acuity, natural or corrected, under bright light of not less than 100 foot-candles. A general-purpose flashlight and a lamp, positioned 18 inches perpendicular to a surface, will illuminate that surface with an intensity of at least 100 foot-candles.

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4. REGULATORY REQUIREMENTS

Not applicable.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.1.1 Sampling and testing. Samples used for testing shall be taken from each lot, as specified by ISO9000, or other commercially accepted practice.

5.1.1.1 Sampling and testing before fill. Before filling each lot, the cleaning compound solvents shall be tested in accordance with table II for appearance, boiling point, chemical purity, impurities, moisture, free chlorine, free fluoride, and non-volatile residue. Sampling for particle matter shall be in accordance with 5.1.1.2.

5.1.1.2 Sampling and testing after fill. After filling each lot, the cleaning compound solvents shall be tested for particle matter as follows:

a. For 1- and 5-gallon containers, the first, last, and every 24th container shall be tested for particle matter in accordance with table II.

b. For 55-gallon containers, the first, last, and every 5th container shall be tested for particle matter in accordance with table II.

5.2 Market acceptability. The following market acceptability criteria are necessary to document the quality of the cleaning compound solvents to be provided by this CID.

5.2.1 The company furnishing the HCFC-225 cleaning compound solvents shall have been producing a product meeting the requirements of this CID for at least 6 months. Also, the company shall have had the HCFC-225 cleaning compound solvents inspected at a government laboratory.

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6. PACKAGING

6.1 Preservation and packing. Preservation and packing shall be as specified in the contract or order.

6.1.1 Marking. Marking shall be as specified in 6.1.1.1.

6.1.1.1 Safety warning labels. Containers shall be marked with a warning label addressing the following information:

a. Extreme conditions of heat (welding, open flame, misuse, or equipment failure) may produce hazardous decomposition products that include hydrogen fluoride. No smoking.

b. Use with adequate ventilation. In case of spills resulting in significant release in a confined space, warn others and leave immediately. Exposure to excessive vapor concentration in confined spaces may cause significant injury.

c. Avoid eye and prolonged skin contact. In case of contact with the eye, rinse immediately with plenty of water, and seek medical advice. Skin contact: flush skin with plenty of water. If irritation persists, call a physician.

d. For additional health and precautionary information, including air-monitoring methodology, refer to the material safety data sheet or contact the manufacturer.

6.1.2 Material safety data sheet (MSDS). A material safety data sheet shall be prepared and submitted in accordance with FED-STD-313.

6.1.3 Delivery options. The HCFC-225 cleaning compound solvents shall be delivered in one-, five-, or fifty-five gallon containers (see 7.2).

7. NOTES

7.1 Intended use. HCFC-225 cleaning compound solvents are intended for precision cleaning applications, and are compatible with gaseous and liquid oxygen when tested in accordance with ASTM-D2512 and ASTM-G72.

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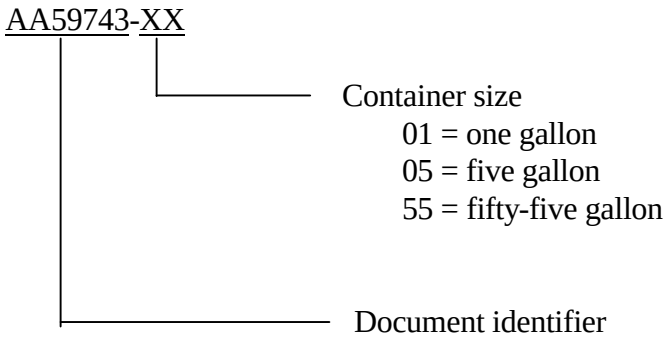
7.2 Part or Identification Number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor. The PIN is constructed as follows:



7.3 Source of documents.

7.3.1 Federal Government activities may obtain copies of FED-STD-313 and DoD adopted non-government standards (NGSs) from established distribution points within their agencies; military activities may obtain these documents from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Other activities may obtain FED-STD-313 from the General Services Administration, Federal Supply Service, Specification Section, Suite 8100, 479 L'Enfant Plaza, SW, Washington, DC 20407 and NGSs from the organization responsible for that document (see 7.3.2 through 7.3.4).

7.3.2 The following documents may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959:

ASTM-D1120	-	Engine Coolants, Boiling Point of. (DoD Adopted)
ASTM-D2109	-	Solvents and their Admixtures, Nonvolatile Matter in Halogenated Organic. (DoD Adopted)
ASTM-D2512	-	Materials with Liquid Oxygen (Impact Sensitivity Threshold and Pass-Fail Techniques), Compatibility of. (DoD Adopted)
ASTM-D3401	-	Water in Halogenated Organic Solvents and their Admixtures. (DoD Adopted)
ASTM-D3443	-	Trichlorotrifluoroethane, Chloride in. (DoD Adopted)
ASTM-D3447	-	Solvents, Purity of Halogenated Organic. (DoD Adopted)
ASTM-F331	-	Residue, Nonvolatile of Halogenated Solvent Extract from Aerospace Components (Using Rotary Flash Evaporator).
ASTM-F661	-	Particle Count and Size Distribution Measurement in Batch Samples for Filter Evaluation Using an Optical Particle Counter, Standard Practice for.
ASTM-G72	-	Liquids and Solids in a High Pressure Oxygen-Enriched Environment, Autogenous Ignition Temperature of.

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7.3.3 The following document may be obtained from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096:

SAE-ARP598	-	Aerospace Microscopic Sizing and Counting of Particulate Contamination for Fluid Power Systems. (DoD Adopted)

7.3.4 The following International Standardization Organization (ISO) document may be obtained from the American National Standards Institute (ANSI), 11 West 42nd Street, New York, NY 10036:

ISO9000	-	Quality Management Systems – Fundamentals and Vocabulary. (DoD Adopted)

7.4 Government laboratory. Manufacturers wanting to have their cleaning compound solvents inspected may do so by contacting Commander, Attn.: Code 4.3.4.1, Building 2188, Naval Air Warfare Center Aircraft Division, Materials Lab, 22347 Cedar Point Road, Unit 6, Patuxent River, MD 20670-1161.

7.5 Purity and cleanliness of container. To conform to requirements of 3.4, all containers, including all caps, should be compatible with the cleaning compound solvent and precleaned before filling. Containers should be filled under environmental conditions and with equipment that will maintain the purity and cleanliness of the solvent. Glass containers should be water flushed and hot-air dried or rinsed with a solvent that is in accordance with this specification. All other containers should also be rinsed with a solvent that is in accordance with this specification before filling.

7.6 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.
b. Product conformance requirements.
c. Packaging requirements.

7.7 Key words.

1,3 Dichloro-1,1,2,2,3, pentafluoropropane
3,3 Dichloro-1,1,1,2,2, pentafluoropropane
HCFC-225

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MILITARY INTERESTS:

Custodians:
Army - EA
Navy - AS

Review activity:
Navy - SH

Preparing Activity:
Navy - AS

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